

Autoencoding Pixies

Amortised Variational Inference
with Graph Convolutions
for Functional Distributional Semantics

Guy Emerson

What I'll Cover...

- Meanings as *functions*, not vectors

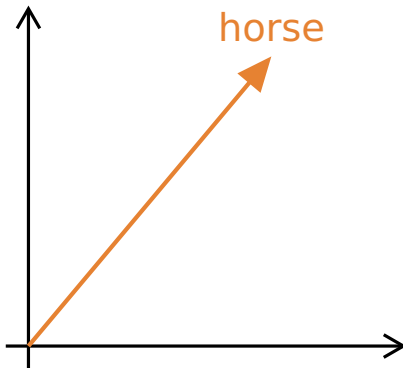
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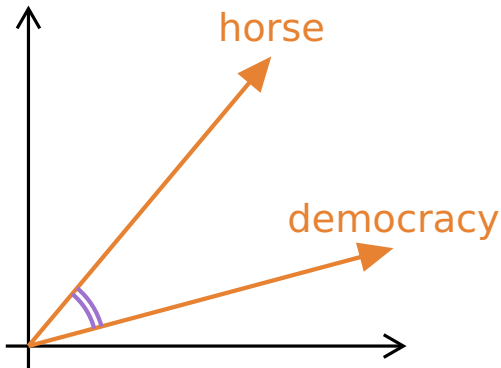
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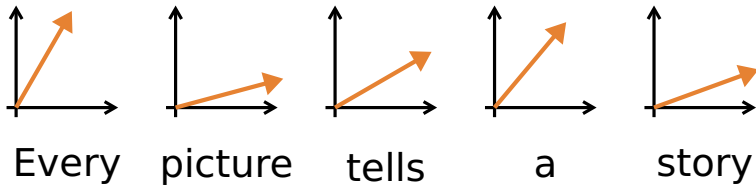
Vector Space Models



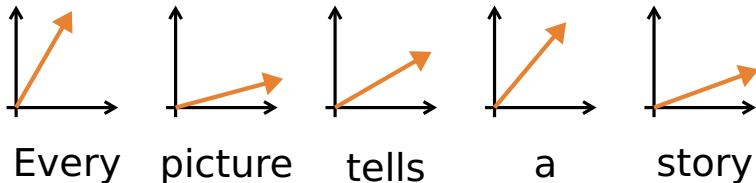
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Vector Space Models for Sentences?

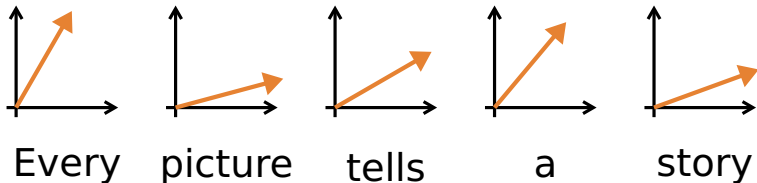


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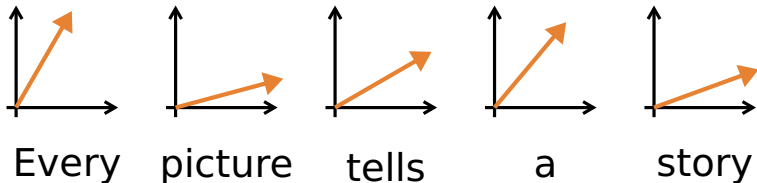
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- Composition? Logic?
- Long history of attempts...
 - See: “What are the Goals of Distributional Semantics?”

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Words are not Entities

- Fundamental distinction between:
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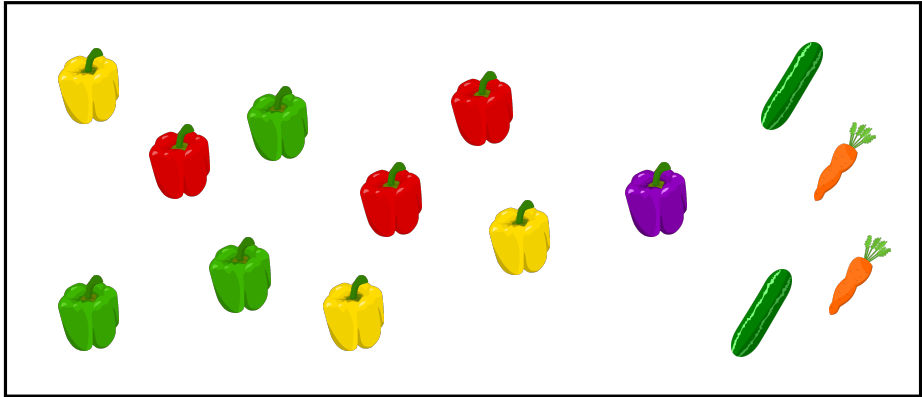
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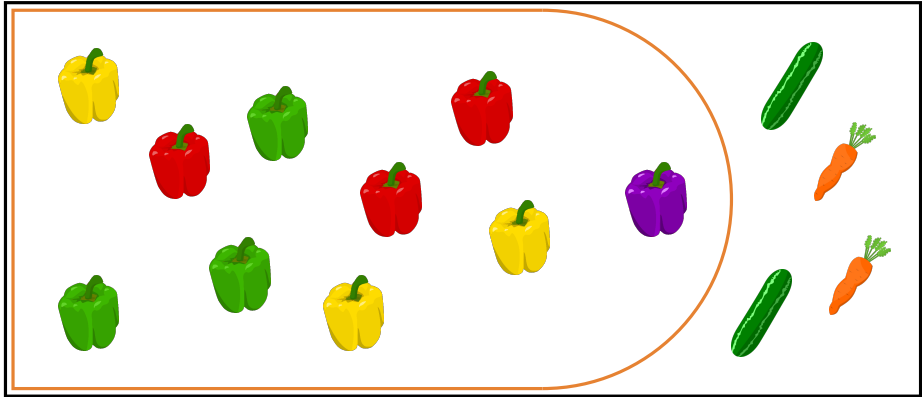
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- NEW: Amortised variational inference
- NEW: Experimental results

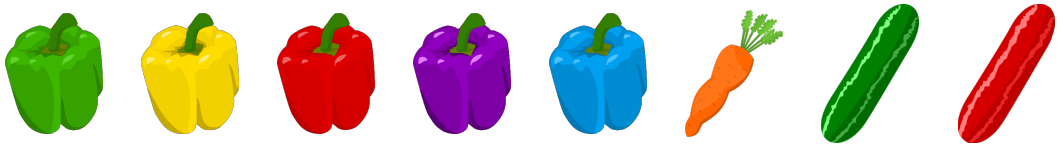
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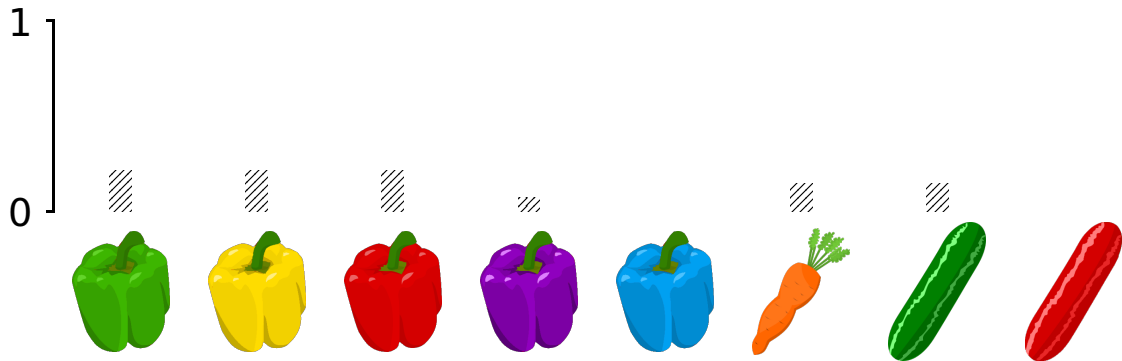
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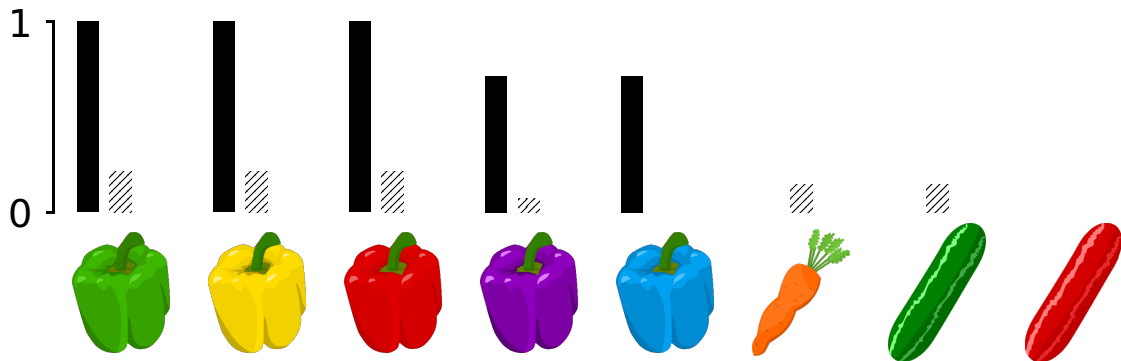
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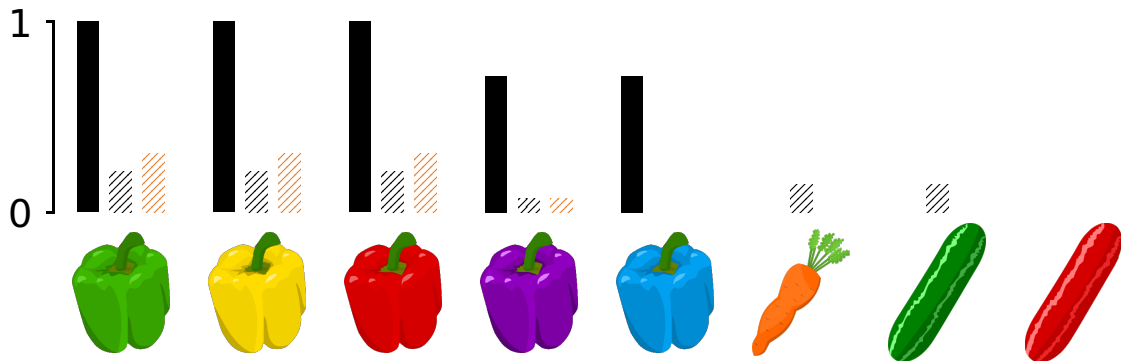
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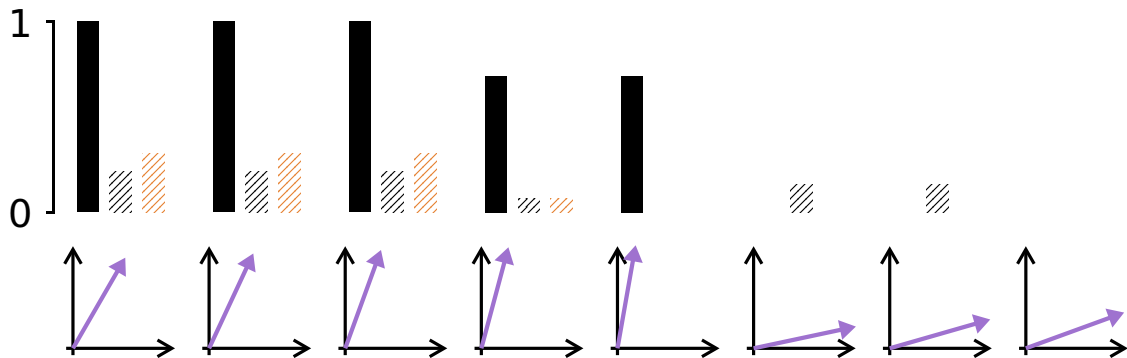
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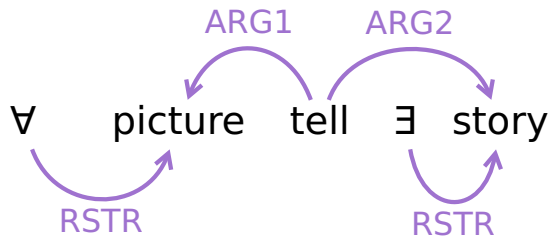
Summary So Far

- Pixie: feature representation of an entity
- Word meanings as functions:
pixie $\mapsto$ probability of truth

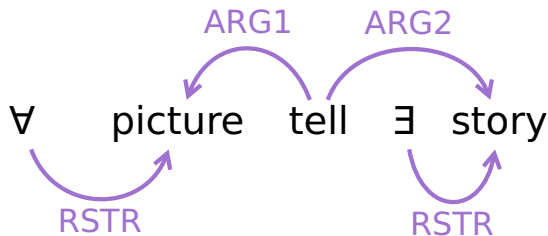
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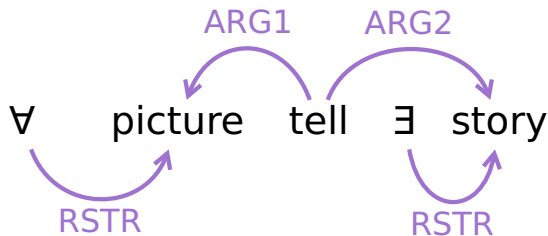
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Functional Distributional Semantics

dog $\xleftarrow{\text{ARG1}}$ chase $\xrightarrow{\text{ARG2}}$ cat

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$$x \xleftarrow{\text{ARG1}} y \xrightarrow{\text{ARG2}} z$$

dog(x) chase(y) cat(z)

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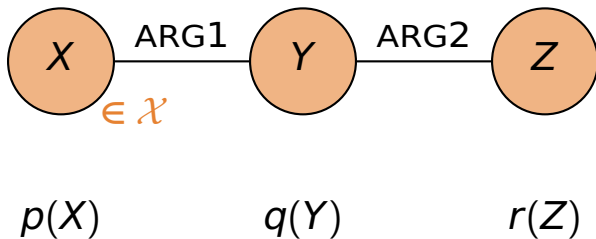
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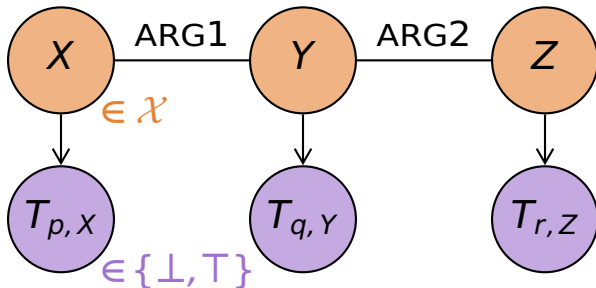
$q(y)$

$r(z)$

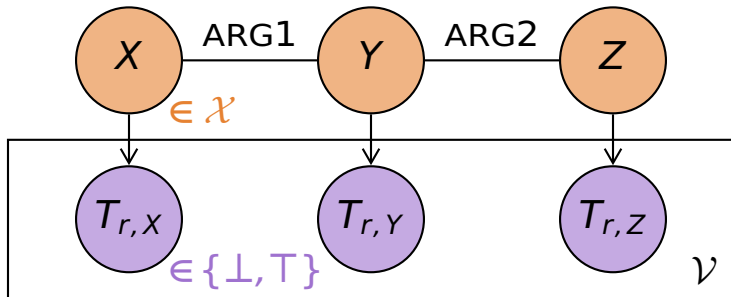
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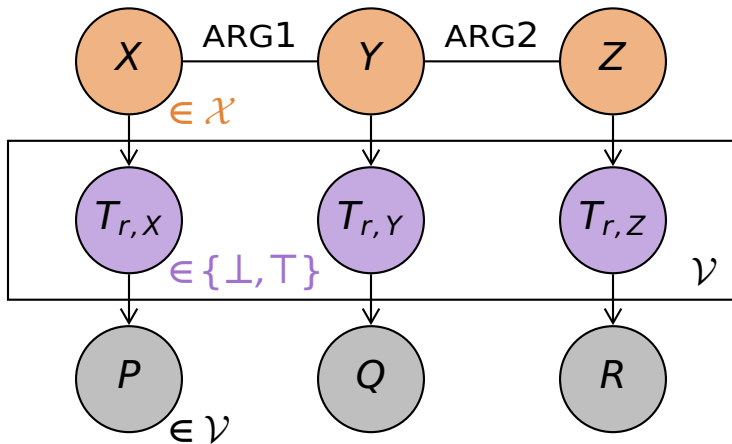
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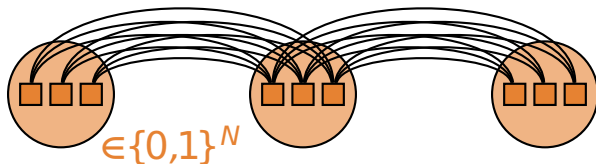
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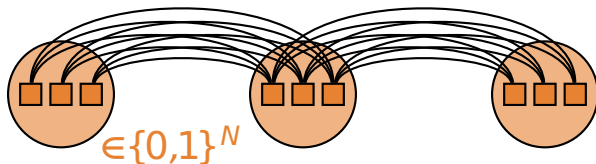


World Model



- Cardinality Restricted Boltzmann Machine
(CaRBM; Swersky et al., 2012)
- $\mathbb{P}(s) \propto \exp(-E(s))$

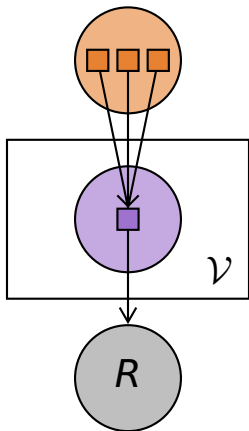
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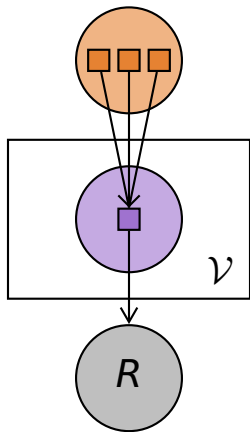
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Lexical Model



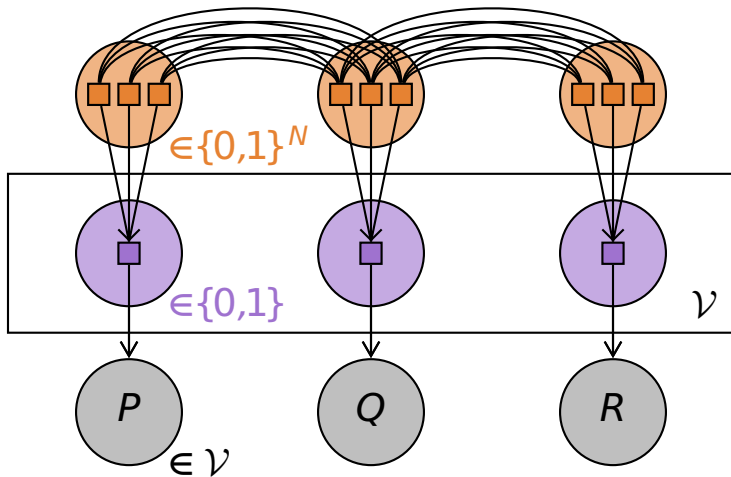
- Feedforward networks
- $t^{(r)}(x) = \sigma(v_i^{(r)} x_i)$

Lexical Model



- Feedforward networks
- $t^{(r)}(x) = \sigma(v_i^{(r)} x_i)$
- $\mathbb{P}(r|x) \propto t^{(r)}(x)$

Functional Distributional Semantics



Gradient Descent

$$\begin{aligned}\frac{\partial}{\partial \theta} \log \mathbb{P}(g) &= \left(\mathbb{E}_{s|g} - \mathbb{E}_s \right) \left[\frac{\partial}{\partial \theta} (-E(s)) \right] \\ &\quad + \mathbb{E}_{s|g} \left[\frac{\partial}{\partial \theta} \log \mathbb{P}(g|s) \right]\end{aligned}$$

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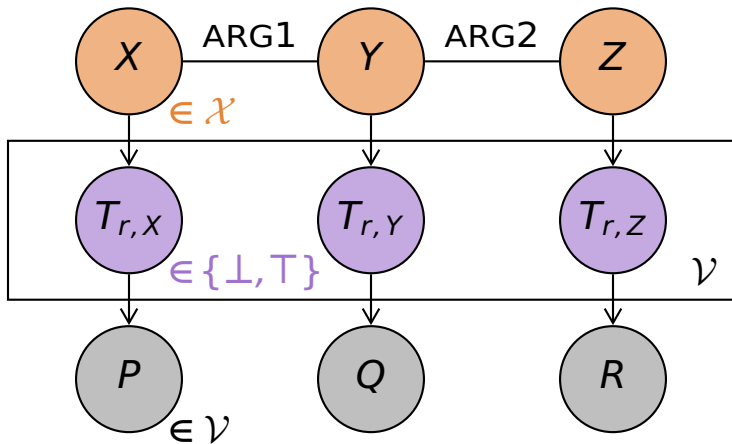
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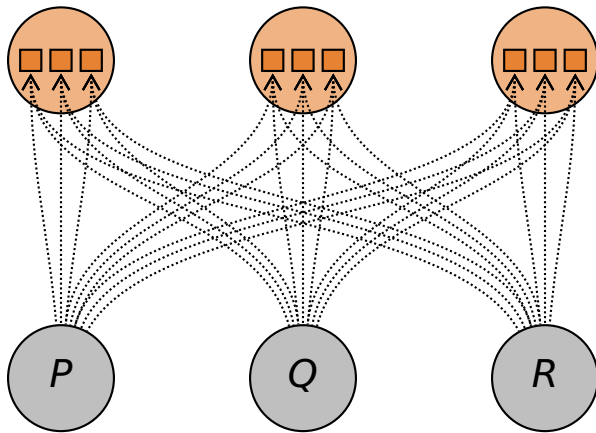
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- Latent variables necessary but inconvenient
- Approximate distribution: variational inference (Jordan et al., 1999; Attias, 2000)

Functional Distributional Semantics



Variational Inference



Amortised Variational Inference

- Variational distribution must be optimised *for each input graph*

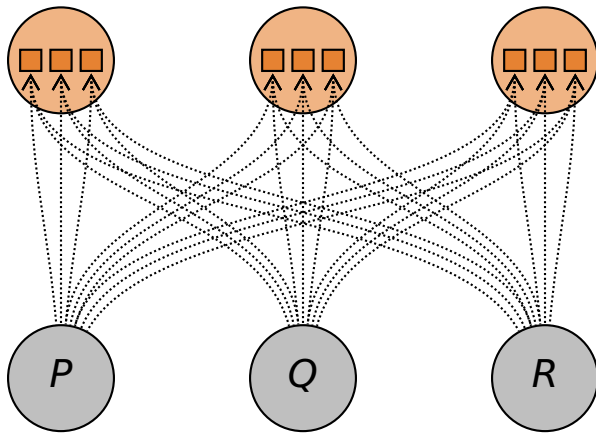
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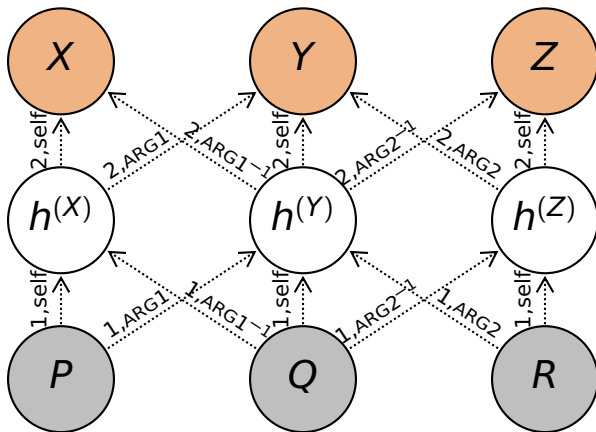
Amortised Variational Inference

- Variational distribution must be optimised *for each input graph*
- Amortisation: train a network to predict the variational distribution (Kingma and Welling, 2014; Rezende et al., 2014; Mnih and Gregor, 2014)
- Input graphs of different topologies: share network weights with graph convolutions (Duvenaud et al., 2015; Marcheggiani and Titov, 2017)

Variational Inference



Amortised Variational Inference



Amortised Variational Inference

$$\begin{aligned}\frac{\partial}{\partial \phi} D(\mathbb{Q}|\mathbb{P}) = & - \frac{\partial}{\partial \phi} \mathbb{E}_{\mathbb{Q}(s)} [\log \mathbb{P}(s)] \\ & - \frac{\partial}{\partial \phi} \mathbb{E}_{\mathbb{Q}(s)} [\log \mathbb{P}(g|s)] \\ & - \frac{\partial}{\partial \phi} H(\mathbb{Q})\end{aligned}$$

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- Latent variables: amortised variational inference

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- Latent variables: amortised variational inference
- Additional details... regularisation, dropout, β -VAE weighting, negative sampling, probit approximation, learning rate, warm start, soft constraints, belief propagation for $\mathbb{E}_s \dots$

Pixie Autoencoder



- Generative model & inference network

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- NLP interest:
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- NLP interest:
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- General ML interest:
 - Efficient inference for latent variables

Training Needs Graphs

- Training needs dependency graphs, not raw text

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- WikiWoods
 - English Wikipedia, parsed into DMRS graphs
 - 31 million graphs (after preprocessing)

Similarity in Context (GS2011)

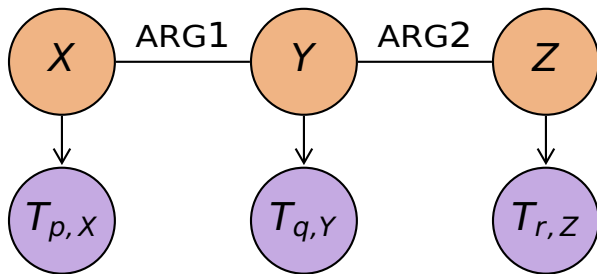
student	write	name
student	spell	name
scholar	write	book
scholar	spell	book

BERT for GS2011

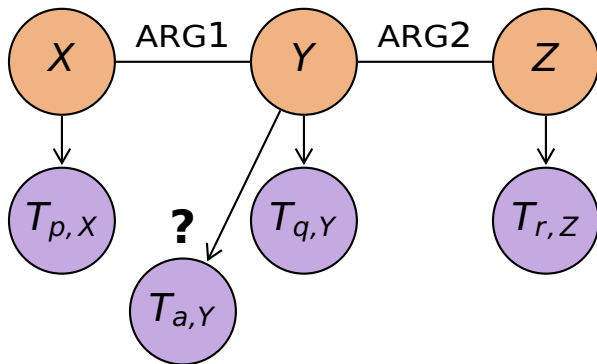
Pseudo-logical form: (employer provide training)

- “an employer **provides** training .”
- “employer **provides** training .”
- “an employer **provides** a training .”
- “a employer **provides** a training .”
- “employers **provide** training .”
- “employers **provide** trainings .”
- “training is **provided** by an employer .”
- “trainings are **provided** by employers .”
- ...

Pixie Autoencoder for GS2011



Pixie Autoencoder for GS2011



$$\mathbb{P}(T_{a,Y} \mid T_{p,X}, T_{q,Y}, T_{r,Z})$$

GS2011 Results

Model	Correlation
Skip-gram (vector addition)	.348
BERT (with tuned template strings)	.446
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- Smaller model, less data, better performance
- More results in the paper!

Summary

- Meanings: functions
- Sentences: graphs
- Inference: graph convolutions
- Logic: useful

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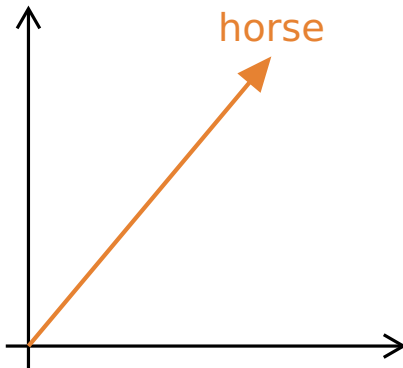
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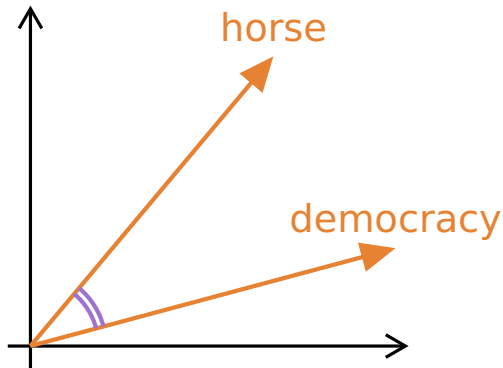
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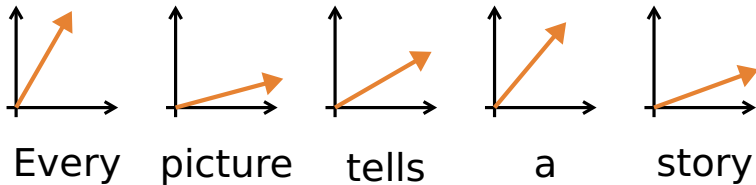
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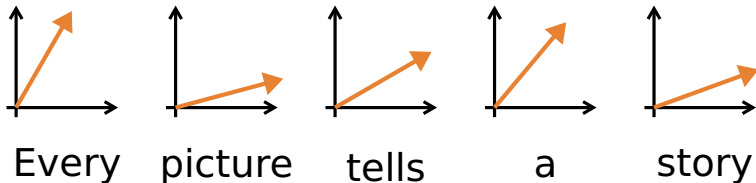
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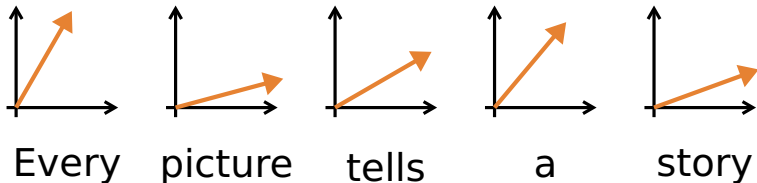


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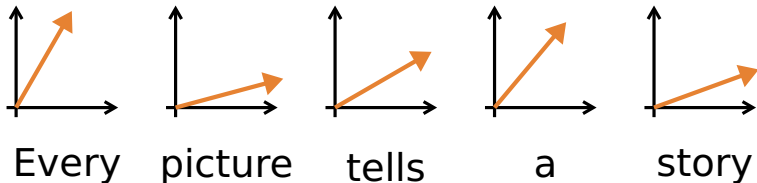
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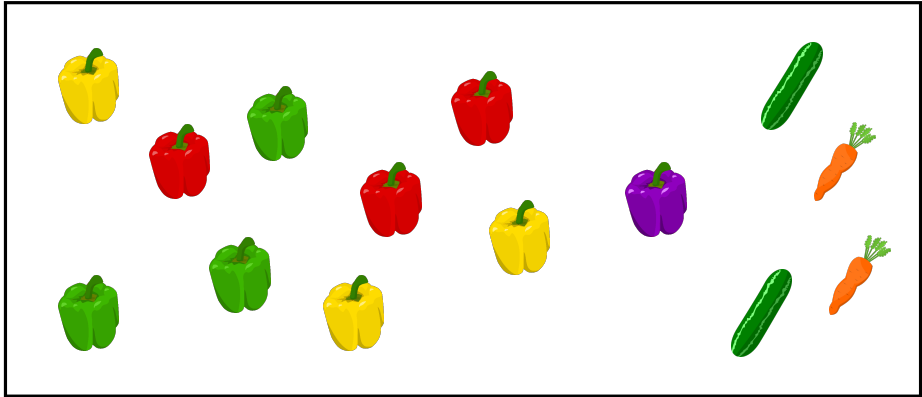
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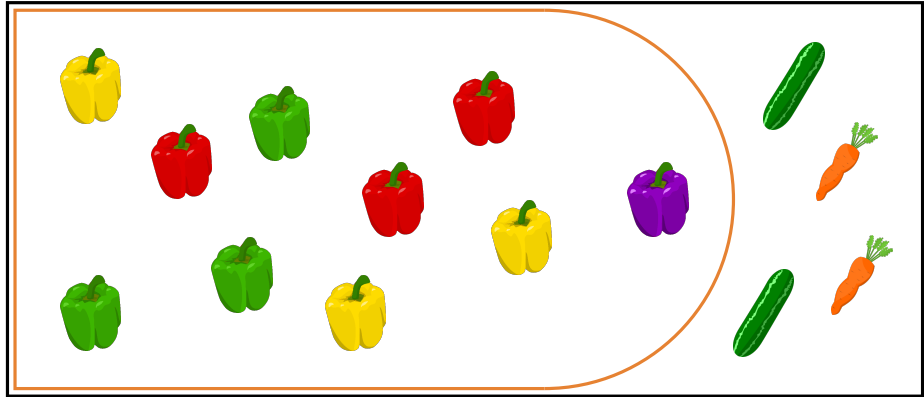
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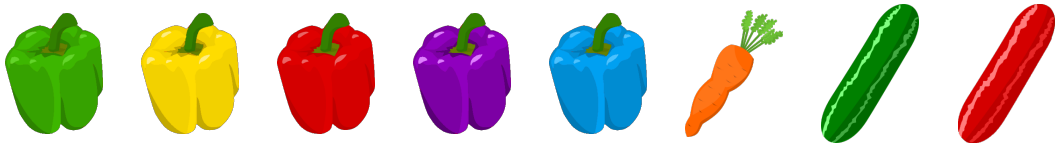
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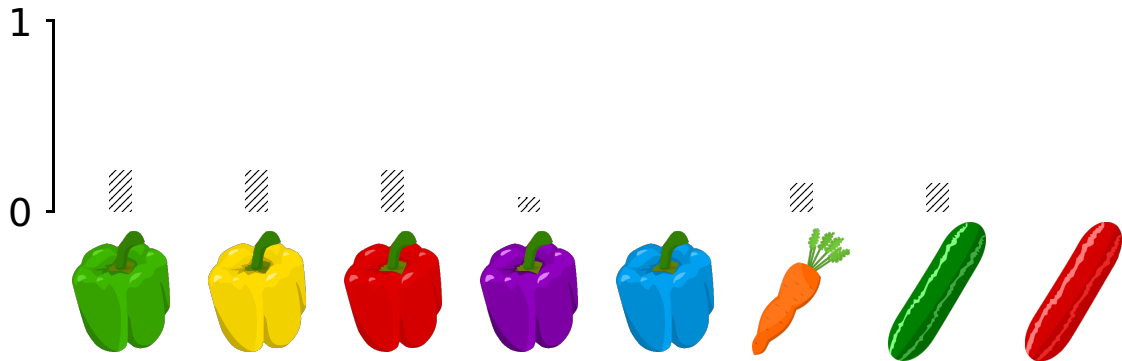
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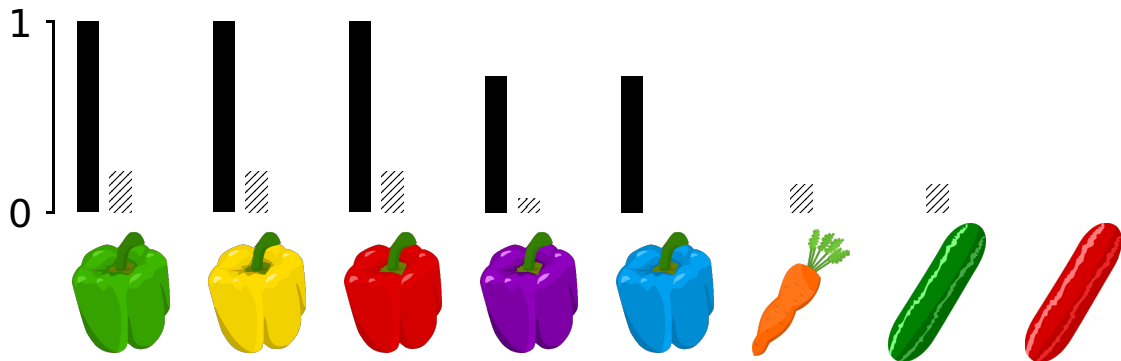
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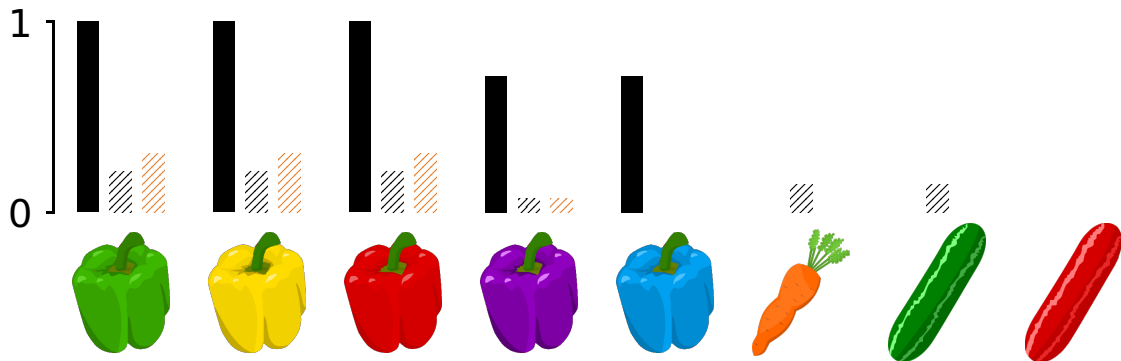
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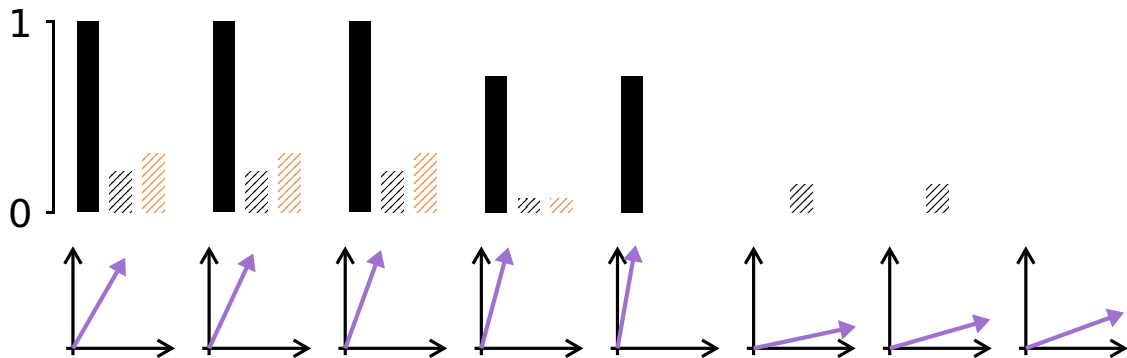
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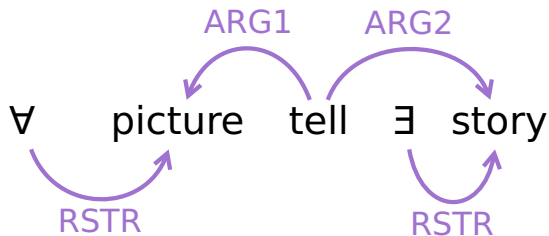
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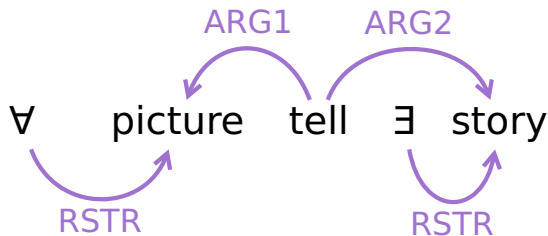
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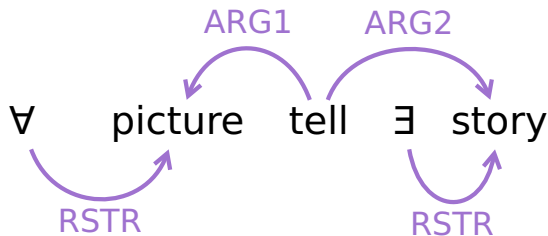


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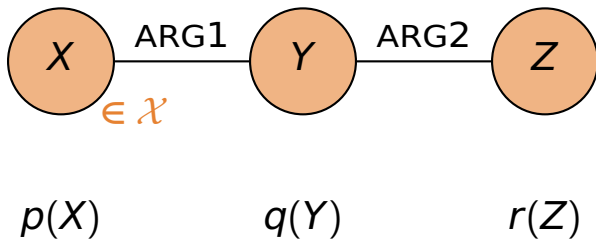
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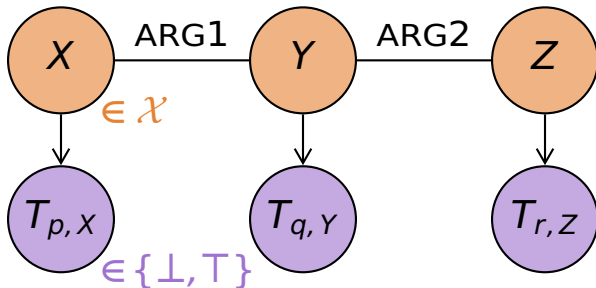
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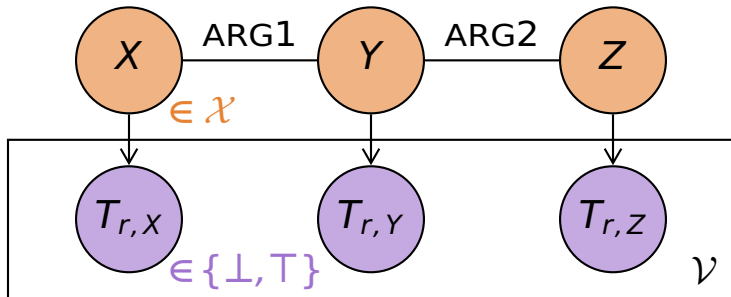
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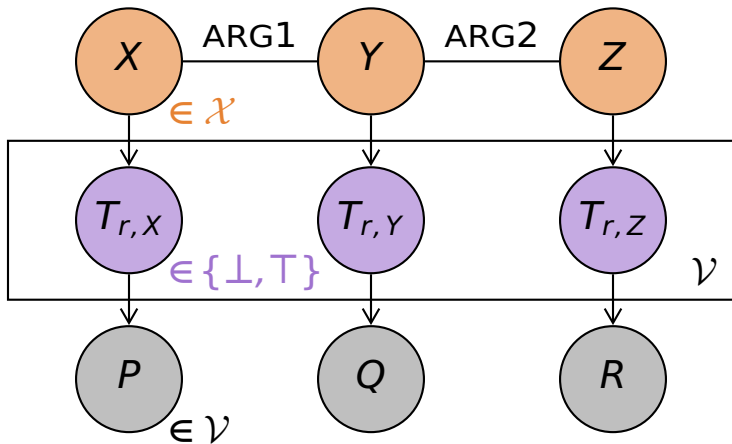
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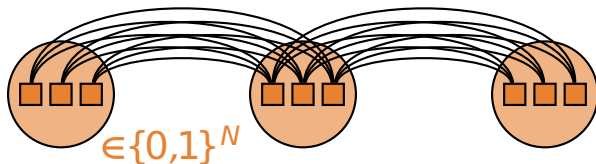
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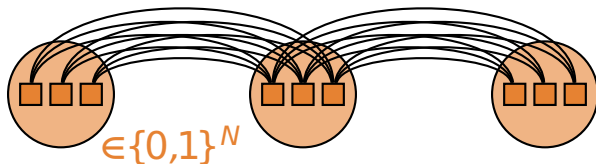


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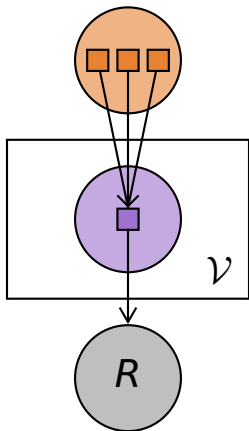
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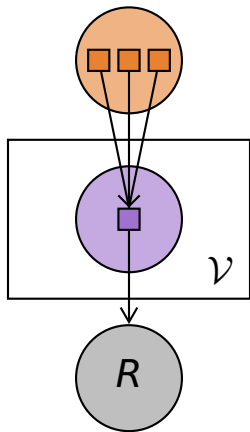
- $\mathbb{P}(s) \propto \exp \left(\sum_{x \xrightarrow{L} y \text{ in } s} w_{ij}^{(L)} x_i y_j \right)$

Lexical Model



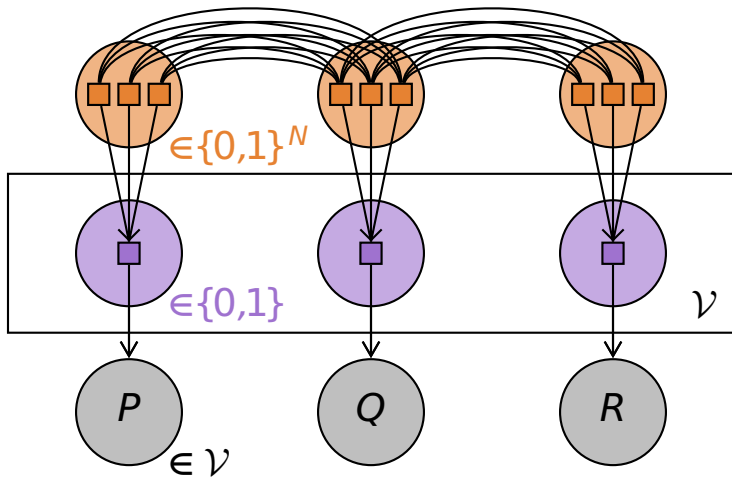
- Feedforward networks
- $t^{(r)}(x) = \sigma(v_i^{(r)} x_i)$

Lexical Model



- Feedforward networks
- $t^{(r)}(x) = \sigma(v_i^{(r)} x_i)$
- $\mathbb{P}(r|x) \propto t^{(r)}(x)$

Functional Distributional Semantics



Gradient Descent

$$\begin{aligned}\frac{\partial}{\partial \theta} \log \mathbb{P}(g) &= \left(\mathbb{E}_{s|g} - \mathbb{E}_s \right) \left[\frac{\partial}{\partial \theta} (-E(s)) \right] \\ &\quad + \mathbb{E}_{s|g} \left[\frac{\partial}{\partial \theta} \log \mathbb{P}(g|s) \right]\end{aligned}$$

Gradient Descent

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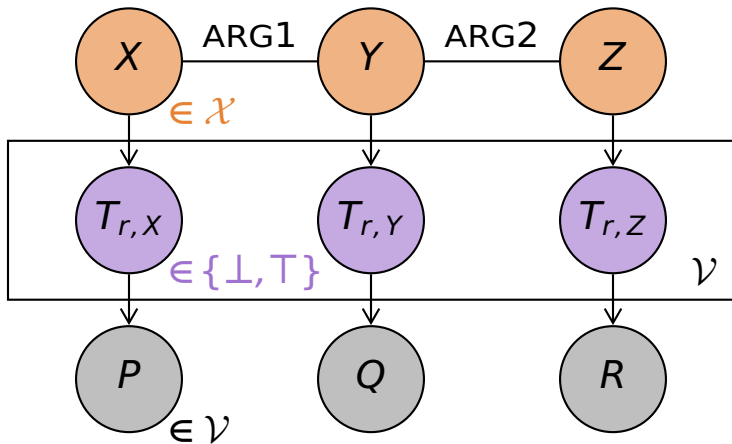
- Latent variables necessary but inconvenient

Gradient Descent

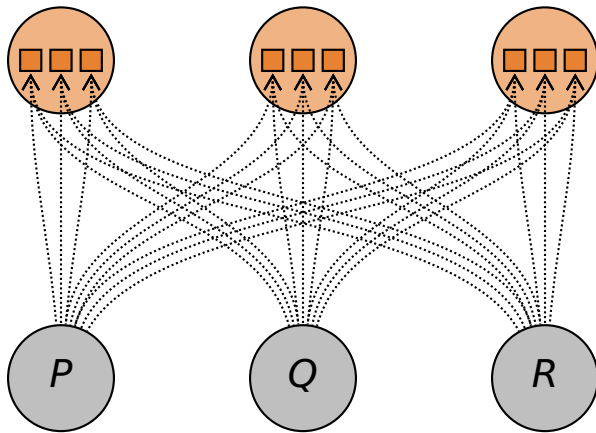
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- Latent variables necessary but inconvenient
- Approximate distribution: variational inference (Jordan et al., 1999; Attias, 2000)

Functional Distributional Semantics



Variational Inference



Amortised Variational Inference

- Variational distribution must be optimised *for each input graph*

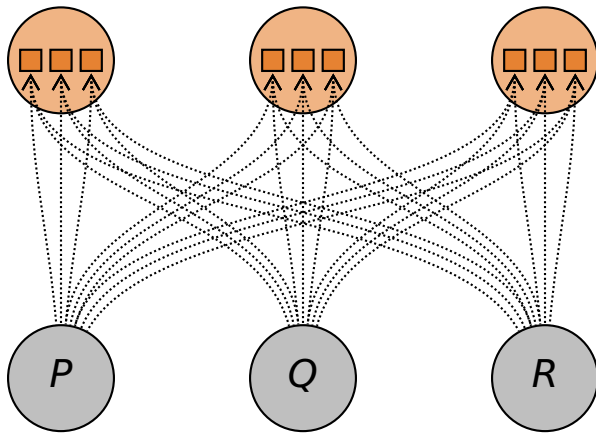
Amortised Variational Inference

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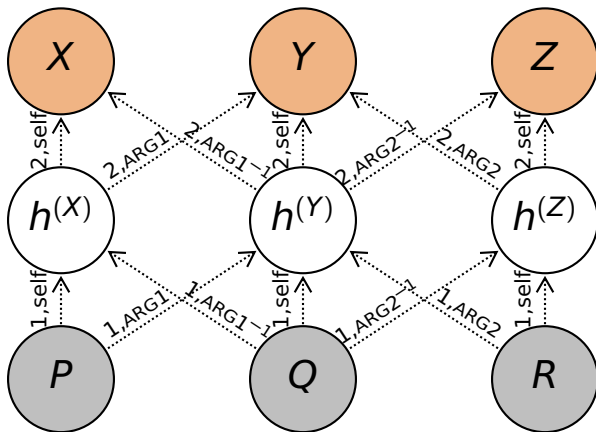
Amortised Variational Inference

- Variational distribution must be optimised *for each input graph*
- Amortisation: train a network to predict the variational distribution (Kingma and Welling, 2014; Rezende et al., 2014; Mnih and Gregor, 2014)
- Input graphs of different topologies: share network weights with graph convolutions (Duvenaud et al., 2015; Marcheggiani and Titov, 2017)

Variational Inference



Amortised Variational Inference



Amortised Variational Inference

$$\begin{aligned}\frac{\partial}{\partial \phi} D(\mathbb{Q}|\mathbb{P}) = & - \frac{\partial}{\partial \phi} \mathbb{E}_{\mathbb{Q}(s)} [\log \mathbb{P}(s)] \\ & - \frac{\partial}{\partial \phi} \mathbb{E}_{\mathbb{Q}(s)} [\log \mathbb{P}(g | s)] \\ & - \frac{\partial}{\partial \phi} H(\mathbb{Q})\end{aligned}$$

Gradient Descent

$$\begin{aligned} \frac{\partial}{\partial \theta} \log \mathbb{P}(g) = & \left(\mathbb{E}_{s|g} - \mathbb{E}_s \right) \left[\frac{\partial}{\partial \theta} (-E(s)) \right] \\ & + \mathbb{E}_{s|g} \left[\frac{\partial}{\partial \theta} \log \mathbb{P}(g|s) \right] \end{aligned}$$

- Latent variables: amortised variational inference

Gradient Descent

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- Latent variables: amortised variational inference
- Additional details... regularisation, dropout, β -VAE weighting, negative sampling, probit approximation, learning rate, warm start, soft constraints, belief propagation for $\mathbb{E}_s \dots$

Pixie Autoencoder

- Generative model & inference network

Pixie Autoencoder

- Generative model & inference network
- NLP interest:
 - Truth-conditional distributional semantics

Pixie Autoencoder

- Generative model & inference network
- NLP interest:
 - Truth-conditional distributional semantics
- General ML interest:
 - Efficient inference for latent variables

Training Needs Graphs

- Training needs dependency graphs, not raw text

Training Needs Graphs

- Training needs dependency graphs, not raw text
- WikiWoods
 - English Wikipedia, parsed into DMRS graphs
 - 31 million graphs (after preprocessing)

Similarity in Context (GS2011)

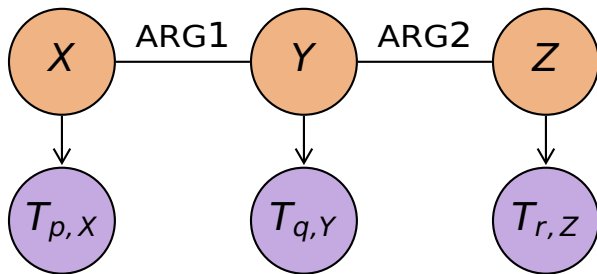
student	write	name
student	spell	name
scholar	write	book
scholar	spell	book

BERT for GS2011

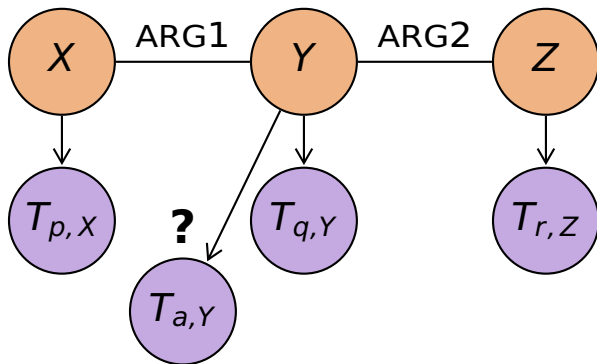
Pseudo-logical form: (employer provide training)

- “an employer **provides** training .”
- “employer **provides** training .”
- “an employer **provides** a training .”
- “a employer **provides** a training .”
- “employers **provide** training .”
- “employers **provide** trainings .”
- “training is **provided** by an employer .”
- “trainings are **provided** by employers .”
- ...

Pixie Autoencoder for GS2011



Pixie Autoencoder for GS2011



$$\mathbb{P}(T_{a,Y} \mid T_{p,X}, T_{q,Y}, T_{r,Z})$$

GS2011 Results

Model	Correlation
Skip-gram (vector addition)	.348
BERT (with tuned template strings)	.446
Pixie Autoencoder	.504

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GS2011 Results

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- Smaller model, less data, better performance
- More results in the paper!

Summary

- Meanings: functions
- Sentences: graphs
- Inference: graph convolutions
- Logic: useful